



# Capital flows under integration: Evidence from China's metropolitan area development planning

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## ABSTRACT

This paper uses the introduction of metropolitan area development planning (MADP) in China to examine the effects of regional integration on capital flow in the context of intranational investment. Using the data of inter-regional capital flows that aggregated by data of listed companies in China from 1999 to 2019, we find that MADP can improve inter-regional investment substantially at the city-pair level. The mechanism analysis indicates that MADP fosters capital flow by diminishing institutional distance, erasing market segmentation, and enhancing transportation infrastructure connectivity. Furthermore, the influence of MADP on capital flows is more significant in cities endowed with robust big data application foundations, lower levels of total factor productivity (TFP), and in city pairs with no administrative borders.

## 1. Introduction

The strategic utilization of regional planning to facilitate the integration and harmonized growth of megacities and their adjoining regions plays a pivotal role in shaping a nation's economic trajectory and development prospects (López-Bazo, Vayá, & Artís, 2004; World Bank, 2013). The resurgence of regionalism in the 20th century has been marked by the adoption of regional planning by local governments to mitigate spatial disparities within regions (Basolo, 2003; Cullingworth, 2003; Mitchell-Weaver, Miller, & Deal, 2000). Metropolitan Area Development Planning (MADP) emerges as an innovative approach that transcends the confines of single administrative entities and geographical boundaries, bridging the gaps inherent in conventional urban system planning constrained by administrative divisions. This paradigm shift constitutes a significant institutional advancement within the urban planning framework of developing nations, infusing vigor into megacities and their peripheries while effectively tackling challenges arising from localized protectionism, administrative decentralization, and fiscal autonomy. Despite its promise, empirical research on MADP and its underlying mechanisms remains limited.

As shown in Fig. 1, the average number of invested firms in city pairs is depicted in the bar chart, from which it is evident that MADP city pairs exhibit a relatively rapid increase in the average number of invested enterprises compared to the non-MADP group. The line graph illustrates the average investment amount, showing that cross-regional investment in cities without MADP experienced a steady growth, while city pairs within the MADP framework demonstrate a notably faster upward trend in cross-regional investment amounts. This paper endeavors to shed light on the substantial role played by MADP in driving inter-regional investments and explore

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the underlying mechanisms that influence capital flows.

This paper assesses the impact and mechanism of MADP on capital flow and allocation at the city pair level. The capital flow data is derived from the matching of affiliated data of publicly listed Chinese companies and city statistical yearbook data. This depicts the capital movement between Chinese cities and reflects the genuine economic linkages brought about by regional integration. The scope and timeframe of metropolitan area development planning are sourced from official documents of provincial Party committees and governments. We employ a Difference-in-Differences (DID) model to ascertain the effect of MADP on inter-regional investment.

The results indicate that the implementation of MADP can improve inter-regional investment significantly at the city-pair level. Mechanism analysis indicates that MADP fosters capital flow by diminishing institutional distance, erasing market segmentation, and enhancing transportation infrastructure connectivity. Heterogeneity analysis further reveals that the influence of MADP on capital flows is more significant in cities endowed with robust big data application foundations, lower levels of total factor productivity (TFP), and in city pairs with no administrative borders.

Compared with the existing literature, the marginal contributions of this paper lie in the following four aspects. First, this paper considers the intensive and extensive margins of regional integration that affect capital flows at the city-pair level. The mainstream literature examines the effect of regional integration on capital flows, and more discussions focus on the effect of intercountry integration on foreign direct investment, mergers and acquisitions, and innovation (Blevins, Moschieri, Pinkham, & Ragozzino, 2016; Moschieri, Ragozzino, & Campa, 2014; Motta & Norman, 1996; Wang, Yeung, Li, & Wang, 2022). However, neither extensive nor intensive margins of capital flows under intranational integration have been addressed. The volume and scale of inter-city enterprise investment obtained from the inter-regional investment data of enterprises in this paper are regarded as the extensive and intense margins, respectively, which indicate to what extent MADP to increase capital flow is effective.

Second, it enriches the discourse surrounding regional integration policies concerning institutional and regulatory disparities arising from boundaries. Within previous literature, discussions on institutional disparities within urban systems can be categorized into two main facets. Firstly, it encompasses governmental favoritism towards cities with higher administrative tiers, involving preferential policies, fiscal bias, trade bias, and capital bias across varying administrative levels. This results in discrepancies in the institutional priority for resource allocation (Desmet & Henderson, 2015; Y. Chen, Henderson, & Cai, 2017). Secondly, it pertains to institutional variations stemming from the existence of boundaries. This encompasses limitations on factor mobility due to localized policy and regulatory incongruities. Such boundary-induced institutional variations influence the feasibility, convenience, and impact of factor connections within administrative borders. This includes systems such as household registration (hukou) and investment promotion policies (Dong, Li, Balland, & Zheng, 2022; Ngai, Pissarides, & Wang, 2019). While a substantial amount of existing literature has scrutinized the effects of resource-related institutional biases stemming from disparities at the city level on the spatial distribution of capital or its mobility, this paper endeavors to explore the integrative role of MADP in influencing boundary-induced institutions and their subsequent impact on capital flow.

Third, we supplemented empirical evidence on the role of government cooperation in driving infrastructure development. A considerable body of literature has explored the economic implications of infrastructure establishment, particularly focusing on whether infrastructure promotes economic differentiation (Baum-Snow, 2007; Baum-Snow, Henderson, Turner, Zhang, & Brandt, 2020). Larger cities tend to gain more resources and agglomeration effects from infrastructure development. As a result, local

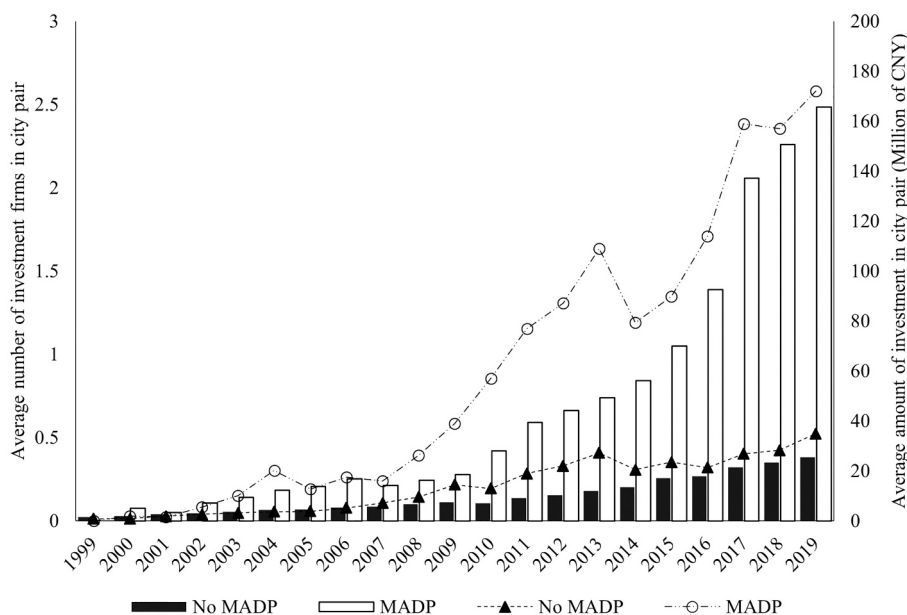


Fig. 1. Evolution of intra-regional investment in city pairs from 1999 to 2019.

Notes: The bar chart depicts the average number of invested firms, while the line chart represents the average amount of investment.

governments often hesitate to support the construction of public transportation infrastructure, maintaining barriers and underdevelopment in border regions. However, within the context of MADP, a significant role exists in fostering cooperation among various local governments in the realm of infrastructure connectivity. In order to optimize resource allocation on a broader scale, MADP actively encourages collaboration among local governments on infrastructure interconnection projects. Through enhancing inter-regional transportation arteries, MADP reduces factor mobility costs and facilitates the optimization of spatial resource allocation across regions. The establishment of transportation infrastructure not only interconnects more cities, forming a network, but also attracts a greater influx of factors, thus promoting the agglomeration economy of metropolitan areas (Redding & Turner, 2015; Xu, Zhang, Zhang, Wang, & Zhang, 2019).

Fourth, there may be endogeneity problems in the MADP. Governments typically plan for metropolitan areas with better industrial development or close investor links. Several studies have employed generalized moment estimation or the DID approach to solve this issue (Alder, Shao, & Zilibotti, 2016; Brühlhart & Mathys, 2008). To reduce endogeneity concerns, this paper proposed a novel IV that incorporates the land gradient (difference of mean slope value), and employed a two-stage least squares (2SLS) technique based on the staggered DID method (Hanley, Li, & Wu, 2022; Nunn & Qian, 2014). The inherent characteristics of natural geographical factors make it difficult to establish a direct link with urban economic development and corporate investment activities (Combes & Gobillon, 2015; Nunn & Puga, 2012). These factors are fundamentally unaffected by urban economic levels and corporate behaviors. Consequently, natural geographical factors can exert an indirect influence on regional capital flows, thus ensuring the exogeneity of this factor.

The remainder of this paper is structured as follows: Section 2 presents the institutional background and theoretical analysis; Section 3 introduces the empirical strategy and data sources; Section 4 presents the empirical analyses; Section 5 presents further analysis; and Section 6 presents conclusions and policy implications.

## 2. Background and hypotheses

### 2.1. Background

The construction of metropolitan areas has required special historical circumstances; thus, further planning and coordination of many regions is still required according to their current growth process (Gagné, Riou, & Thisse, 2016). For instance, under the guidance of government planning, the metropolitan areas of London and Tokyo have undergone several significant developments. Therefore, development planning for metropolitan areas in China is necessary. On the one hand, the decentralization reform in China has increased the authority and autonomy of local governments, making intergovernmental interactions challenging for regional integration. On the other hand, regional heterogeneity makes it difficult to replicate sophisticated demonstration experiences during local development, forcing local governments to pursue collaboration and integration on a greater scale and investigate comprehensive internal solutions (Asheim, Boschma, & Cooke, 2011).

Once the *Seventh Five-Year Plan* transferred the focus of reform from rural to urban areas, China's urbanization trajectory changed from "urbanization dominated by small and medium-sized cities and strict control of the size of large cities" in the *Eighth Five-Year Plan* in 1991 to "coordinated development of large and medium-sized cities" in the *Ninth Five-Year Plan* in 1996, paving the way for the emergence of metropolitan areas. Since the turn of the century, China's provinces and cities have actively responded to the call of the Central Committee's *Ninth Five-Year Plan*, have issued plans to cultivate metropolitan areas, and have established metropolitan area leading groups as coordination mechanisms or regular joint meetings as coordination platforms. Further, 31 provinces (cities) have released local provincial metropolitan area development planning by 2018,<sup>1</sup> and 127 cities have been included in the planning. Fig. 2 shows that by 2018 metropolitan area development planning had been established in most regions throughout the country.

The MADP framework encompasses five key elements: spatial integration, economic integration, infrastructure development, social public services, and governance and policy coordination. The first key element, spatial integration, focuses on delineating the metropolitan area's spatial boundaries and layout. This includes identifying core cities and peripheral regions earmarked for targeted development. A concerted effort is made to seamlessly integrate these areas, optimizing transportation and connectivity across the agglomeration.

The second crucial aspect, economic integration, involves fostering balanced growth across industries, value chains, and innovation hubs within different cities. This strategic approach aims to capitalize on economies of scale and enhance the overall competitiveness of the metropolitan area.

The third fundamental aspect, infrastructure development, entails strategic planning and coordination of vital transportation, communication, and public service infrastructure. This collective endeavor aims to ensure the smooth and efficient functioning of the entire agglomeration.

The fourth pivotal element, social public services, focuses on enhancing essential amenities such as housing, healthcare, education, and cultural facilities. This augmentation is designed to elevate the overall quality of life for residents throughout the agglomeration.

The final critical element involves governance and policy coordination, encompassing the establishment of efficient governance mechanisms and harmonization of policies across diverse administrative units. This strategic approach ensures the effective implementation of agglomeration planning objectives.

<sup>1</sup> In 2019, the State council of China issued the *Guiding Opinions on the Cultivation and Development of Modern Metropolitan Areas*, officially launched the next round of the national construction of metropolitan areas, and approved the metropolitan areas development planning at the central level, based on 20 years of experience in metropolitan area development planning at the province-level.

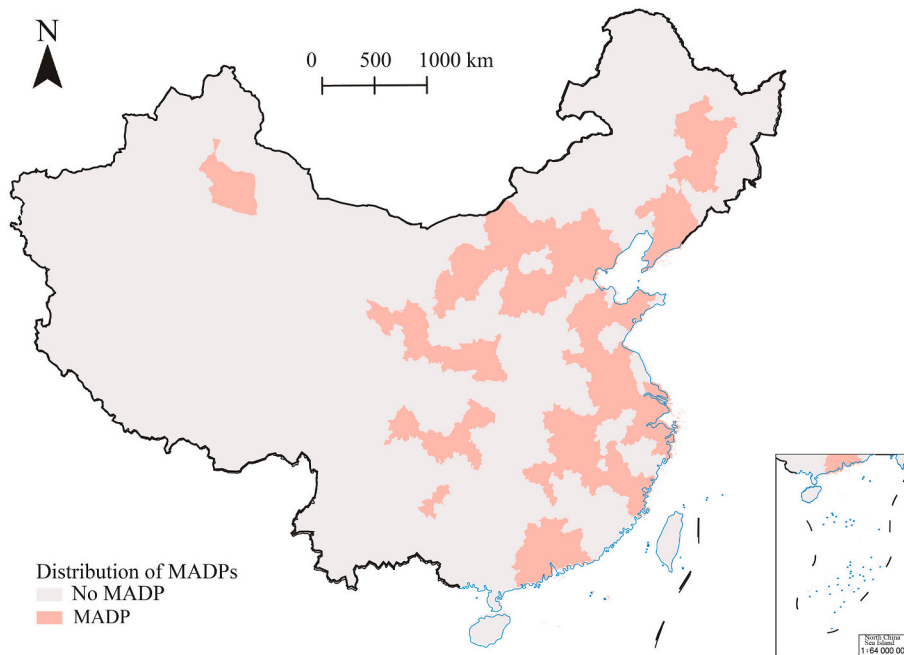


Fig. 2. Location of metropolitan area development planning.

Regarding similarities and differences between different planning approaches, while the core objectives of MADP are generally similar across regions, the specific strategies and emphasis may vary based on local contexts and governance structure. On the one hand, Some MADP may focus more on technology-driven innovation (i.e., Shanghai MADP), while others may prioritize environmental conservation (i.e., Wuhan MADP) and cultural heritage (i.e., Xi'an MADP). On the other hand, differences in political systems and administrative structures may influence the level of policy coordination and implementation effectiveness. While most metropolitan areas establish coordination mechanisms through joint meetings, more advanced regions like Shanghai have ventured to establish integrated pilot zones along boundaries to facilitate multi-party management and knowledge exchange.

## 2.2. Theoretical analysis

The integration strategy serves as a spatial policy that effectively mitigates regional disparities (World Bank, 2009). Regional integration holds the potential to facilitate capital flows among nations, with investment creation and transfer effects observable from the vantage point of international trade theory (Balassa, 1961; Kindleberger, 1966). Davies, Norbäck, and Tekin-Koru (2009) notes that institutional integration can amplify the investment scales of transnational corporations at the intensive margin, although its effect at the extensive margin remains inconsequential. In accordance with existing literatures, integration has the capacity to enlarge market size, diminish transaction costs through inter-regional coordination, enhance investment efficiency, and augment inter-regional capital flows (Baldwin, Forslid, & Haaland, 1995; Neary, 2002).

As an institutional innovation within the realm of regional integration, MADP actively promotes capital flows by diminishing institutional distance, erasing market segmentation, and enhancing transportation infrastructure connectivity.

First, MADP can reduce the institutional distance between cities to improve the capital flow. Because of regional administrative boundaries and disparities in local political resources, inconsistencies in the local system hinder inter-regional enterprise capital movement or investment (Boisot & Meyer, 2008). Regional administrative boundaries result in distinct boundary systems, including policies and regulations, within administrative regions. Additionally, under the jurisdiction of the administrative system, disparities in the supply of urban factors and governance competencies result in an institutional gap (He, Li, & Chen, 2019). Efficient regional integration planning plays a crucial role in promoting institutional cohesion, thereby creating favorable conditions for seamless regional capital mobility (Venables, 1999). This implies that agreements promoting regional cooperation and coordination, such as MADPs, impact inter-regional investments by minimizing the institutional distance between regions. MADP achieves this by coordinating and unifying policies related to investment attraction, environmental regulations, administrative approval systems, and other institutional factors that impact the business environment. It also standardizes market entry for business mobility, thereby reducing institutional friction arising from disparities in capital flow.

Second, MADP reduces market segmentation in order to reduce trade barriers and increase capital flow. Inter-regional investments are costly for Chinese enterprises. Tight competition between local governments to entice businesses to reinvest has resulted in significant municipal protectionism and market segmentation (Young, 2000). Most studies consider local protectionism and regional market segmentation increase the trade costs of enterprises' interregional operations (Pan & Yu, 2011; Ye & Zhang, 2017). Changes in

competitiveness and cooperation between local governments influence the degree of market segmentation between regions as well as friction and resistance to the movement of capital, labor, and other factors (Siegel, Licht, & Schwartz, 2011). MADP shifts local governments from competition to collaboration, guiding regions to establish mutually beneficial relationships in industries, technologies, and public services. This fosters deeper cooperation, reducing trade friction and promoting market integration. Enterprises are drawn to metropolitan areas for easier and more affordable access to larger markets (Giroud & Mueller, 2015; Giroud & Rauh, 2019; Singh, 2005).

Thirdly, MADP has propelled collaborative inter-regional infrastructure development, reducing the spatial mobility costs of factors and mitigating information frictions, thereby facilitating cross-regional investments among cities. However, the presence of boundary effects, particularly concerning infrastructure connectivity, has led to suboptimal inter-regional and especially inter-border connectivity. Nonetheless, a key initiative within MADP involves fostering the implementation of public transportation infrastructure projects through collaborative efforts among local governments. Within the realm of inter-regional development, the introduction of infrastructure projects often engenders a phenomenon known as the “backwash effect” or “polarization effect,” resulting in differing governmental stances when it comes to promoting cooperative infrastructure endeavors (Baum-Snow, 2007; Baum-Snow et al., 2020). In response, MADP takes steps to enhance government cooperation, driving the interconnection of transportation arteries between regions, thereby reducing factor mobility costs and facilitating the optimization of spatial resource allocation across regions (Redding & Turner, 2015). This approach not only contributes to fostering the agglomeration economy within metropolitan areas but also plays a constructive role in nurturing potential metropolitan areas, particularly those contiguous to core urban districts. This phenomenon spurs the emergence of a horizontal and polycentric city network, effectively promoting regional integration (Xu et al., 2019).

### 3. Research design

#### 3.1. Data sources and key variables

To address these issues, we constructed a novel comprehensive database of annual intercity investment data on capital flows using data on listed companies' affiliated enterprises 1F from CSMAR from 1999 to 2019. Additionally, we matched the office address data of the enterprises with city-level data from the *China City Statistical Yearbook* to capture local characteristics. Data on information on MADP was sourced from metropolitan area planning documents issued by provincial governments. A total of 127 cities were successively incorporated into 29 MADPs, resulting in a combined  $297 \times 296$  city pairs. After excluding missing values, the dataset encompassed 1,507,772 observations.

##### 3.1.1. Inter-regional investment

Data pertaining to inter-regional investments by listed companies were collected from the CSMAR databases. Information regarding the locations of listed companies' headquarters and their affiliated entities was extracted from their annual reports, encompassing details such as securities codes, locations, and investment amounts. Once the affiliated entities were identified, we calculated the number of affiliates and aggregated the investment values from each parent company based on the respective locations of the affiliates and headquarters. This process allowed us to determine the investment amount and count for each city pair, thereby quantifying inter-regional investment using both extensive and intensive margins of inter-city investment by listed companies (Davies et al., 2009).

The extensive margin (denoted as  $fnm_{ijt}$ ) represents the total count of listed companies unilaterally investing in their affiliates from  $city_i$  to  $city_j$ . Meanwhile, the intensive margin (denoted as  $cap_{ijt}$ ) is quantified by the logarithm of the aggregate value of investments made by listed companies from  $city_i$  to  $city_j$ .

##### 3.1.2. Metropolitan planning area identification

This paper utilizes manual information collection to identify the scope of MADP, following the steps below:

- (1) We manually sorted the scope of metropolitan area development planning in prefecture-level cities from 1999 to 2019. The scope of MADP is derived from two primary sources. Firstly, we conducted manual searches of province government documents to gather relevant information. Secondly, we used the China National Knowledge Infrastructure (CNKI) platform to query keywords related to metropolitan areas development planning (or “metropolitan area,” “city circle”). Through this process, we verified the implementation scope and timeframe of these policies, ensuring accuracy and enhancing the reliability of the data.
- (2) According to the city where the company's office is located ( $city_i$ ) and the city where the affiliates are located ( $city_j$ ) as the source and destination city of investment, respectively, city-pair level ( $city_{ij}$ ) data are constructed.
- (3) The policy dummy variable  $madp_{ijt}$  is set according to the scope of MADP announced by the provincial (municipal) government from 1999 to 2019. If the city pair  $city_{ij}$  is included in the scope of the metropolitan area in the year  $t$ , the value of  $madp_{ijt}$  is 1; otherwise, it is 0.

##### 3.1.3. Control variables

Building upon existing studies, the following control variables for destination and origin cities were selected: GDP( $\ln GDP$ ), population( $\ln pop$ ), the proportion of the secondary industry( $second$ ), the proportion of the tertiary industry( $third$ ), urbanization rate ( $urbanrate$ ).

### 3.2. Research methods

This paper employs the Difference-in-Differences (DID) method to analyze the impact of the metropolitan area planning policy on inter-regional investment. To achieve this, we conduct regression analyses in the following form:

$$Y_{ijt} = \beta_1 + \beta_2 madp_{ijt} + \beta_3 Control_{it} + \beta_4 Control_{jt} + \lambda_{it} + \mu_{jt} + \varepsilon_{ijt} \quad (1)$$

where  $i$  represents the city of capital outflow (the location of the listed company),  $j$  represents the city of capital inflow (the location of the affiliated enterprise), and  $t$  represents the year. We have two measures for the dependent variable,  $Y_{ijt}$  (Bo, 2020; Davies et al., 2009; Giroud, 2013). The first is  $lnfnm_{ijt}$  the logarithm of sum of the number of all affiliated companies at the city-pair level  $fnm_{ijt}$  (we use the logarithmic value, ) reflecting the extensive margin of capital flow of the city pair ( $i, j$ ). The second variable is  $cap_{ijt}$ , the total investment scale of listed companies to affiliated companies in city pair ( $i, j$ ), which reflects the intensive margin of investment scale in cities, and we also use the logarithmic value  $lcap_{ijt}$  in the model.

The core independent variable in our analysis is the MADP denoted as  $madp_{ijt}$ . It takes a value of 1 if there is an investment outflow from  $city_i$  and an inflow to  $city_j$  within year  $t$  within the scope of the MADPs, and 0 otherwise. Furthermore, the control group comprises pairs of cities that have not yet been included within the scope of the MADPs.

Additionally, *control variables* is a series of economic variables for the origin and destination cities. To remove the cross-section and time-series correlation that result from an omitted variable bias, time-variant origin and destination city dummy variables,  $\lambda_{it}$  and  $\mu_{jt}$  are controlled.  $\varepsilon_{ijt}$  is a random error term, and standard errors are clustered at the city-pair level. Table 1 presents the definition and statistical descriptions of the main variables.

## 4. Empirical results

### 4.1. Baseline estimates

In this section, we apply the specification in Eq. (1) to examine inter-regional investment at city-pair level, i.e., the impact of MADP on investment of the Chinese listed companies between  $city_i$  and  $city_j$ . We estimate Eq. (1), and the empirical results are reported in Table 2. In Column (1,2) of Table 2, we present the estimates results without control variables. Column (3,4) adds the control variables. It can be seen that whether or not the control variables are included, the estimated coefficients of  $madp$  are significantly positive. In Columns (3), we examine the effect of MADP on extensive margin, i.e., the affiliates number of Chinese listed companies between  $city_i$  and  $city_j$ , with an increase of about  $(e^{0.0755} - 1) \times 100\% = 7.84\%$ . In columns (4), we test the effect of MADP on intensive margin, which is the total amount of investment for each city-pair. MADP can also significantly increase total amount of inter-regional investment by  $(e^{0.5936} - 1) \times 100\% = 81.05\%$ . This result indicates that, after the implementation of the MADP between city  $i$  and  $j$  compared with other city-pairs, the total firm number and total amount of inter-regional investment increased significantly.

**Table 1**  
Summary of variables.

| Variables                      | Definition                                                                                                                     | N         | Mean    | S.D.    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------|---------|---------|
| Dependent variables            |                                                                                                                                |           |         |         |
| <i>lnfnm</i>                   | Logarithm of total number of listed companies' affiliates after adding 1 for the city pair( $i$ - $j$ ) in the year $t$        | 1,507,772 | 0.0666  | 0.3161  |
| <i>lncap</i>                   | Logarithm of total amount of listed firms invest in affiliations after adding 1 for the city pair( $i$ - $j$ ) in the year $t$ | 1,507,761 | 0.5563  | 3.2663  |
| Independent variable           |                                                                                                                                |           |         |         |
| <i>madp</i>                    | Dummy for whether city pair is in MADP                                                                                         | 1,507,772 | 0.0845  | 0.2782  |
| Control variables for $city_i$ |                                                                                                                                |           |         |         |
| <i>lngdp</i>                   | $\ln(GDP_i)$                                                                                                                   | 1,507,772 | 6.6177  | 1.1521  |
| <i>lnpop</i>                   | $\ln(popdensity_i)$                                                                                                            | 1,507,772 | 5.8519  | 0.6986  |
| <i>second</i>                  | the proportion of the secondary industry                                                                                       | 1,507,772 | 47.1371 | 11.2366 |
| <i>third</i>                   | the proportion of the tertiary industry                                                                                        | 1,507,772 | 37.8307 | 9.2001  |
| <i>urbanrate</i>               | urbanization rate                                                                                                              | 1,507,772 | 0.0832  | 0.0935  |
| Control variables for $city_j$ |                                                                                                                                |           |         |         |
| <i>lngdp</i>                   | $\ln(GDP_j)$                                                                                                                   | 1,507,772 | 6.6177  | 1.1521  |
| <i>lnpop</i>                   | $\ln(popdensity_j)$                                                                                                            | 1,507,772 | 5.8519  | 0.6986  |
| <i>second</i>                  | the proportion of the secondary industry                                                                                       | 1,507,772 | 47.1371 | 11.2366 |
| <i>third</i>                   | the proportion of the tertiary industry                                                                                        | 1,507,772 | 37.8307 | 9.2001  |
| <i>urbanrate</i>               | urbanization rate                                                                                                              | 1,507,772 | 0.0832  | 0.0935  |



**Table 2**  
Baseline estimates results.

| Variables                         | (1)                   | (2)                   | (3)                   | (4)                   |
|-----------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                   | <i>lnfnm</i>          | <i>lnicap</i>         | <i>lnfnm</i>          | <i>lnicap</i>         |
| <i>madp</i>                       | 0.0858***<br>(0.0224) | 0.6624***<br>(0.1599) | 0.0755***<br>(0.0212) | 0.5936***<br>(0.1542) |
| <i>_cons</i>                      | 0.0582***<br>(0.0018) | 0.4737***<br>(0.0128) | 0.0602***<br>(0.0018) | 0.5062***<br>(0.0130) |
| Control variables                 | NO                    | NO                    | YES                   | YES                   |
| <i>City<sub>i</sub> × Year FE</i> | YES                   | YES                   | YES                   | YES                   |
| <i>City<sub>j</sub> × Year FE</i> | YES                   | YES                   | YES                   | YES                   |
| <i>N</i>                          | 1,846,152             | 1,846,137             | 1,507,772             | 1,507,761             |
| Adj. <i>R</i> <sup>2</sup>        | 0.3642                | 0.1861                | 0.3531                | 0.1792                |

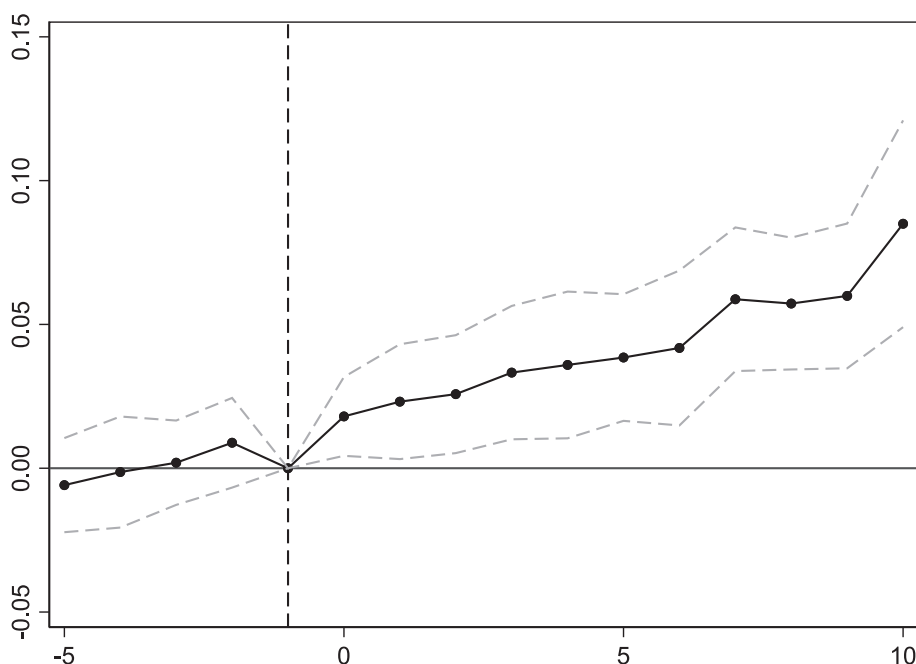
Note: (i) Robust standard errors, clustered at city pair and year level, are reported in brackets. (ii) \*\*\**p* < 0.01, \*\**p* < 0.05, \**p* < 0.1.

#### 4.2. Parallel trend test

The prerequisite for employing the DID (Difference-in-Differences) method is that the treatment and control groups exhibit comparable trends prior to the policy intervention. In our study, we examine whether the patterns of inter-regional investment in the treatment and control groups displayed a shared trend leading up to the implementation of MADP. To assess this, we employed an event study approach commonly utilized in existing literature, conducting a parallel trend analysis (Lu, Tao, & Zhu, 2017). Specifically, we formulated the following regression equation:

$$\ln fnm_{ijt} = \beta_0 + \sum_{k=-5, k \neq -1}^{10+} \alpha_k D_{ijt}^k + \beta_2 Control_{it} + \beta_3 Control_{jt} + \lambda_{it} + \mu_{jt} + \epsilon_{ijt} \quad (2)$$

where  $D_{ijt}^k$  is a series of event dummy variables;  $d_{ijt}$  denotes the year when city pair (*i, j*) was first included in the MADP in year *t*. If  $t - d_{ijt} \leq -5$ ,  $t - d_{ijt} = k$  ( $k = -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$ ), and  $t - d_{ijt} \geq 10$ , then  $D_{ijt}^k = 1$ , otherwise 0. Fig. 3 illustrates that the coefficients of all dummy variables lacked significance prior to the adoption of MADP. These findings indicate a positive influence of MADP on inter-regional investments. Moreover, there was no significant difference between the treatment and control groups prior to the policy intervention.



**Fig. 3.** Parallel trend test of MADP.

Note: The dashed lines represent the 95% confidence interval of the estimated coefficients.

### 4.3. Placebo test

Considering the endogeneity problem caused by omitted and random variables, we employed [Chen, Li, and Lu \(2018\)](#)'s placebo test to randomize the policy shocks. This process was repeated 1000 times using Monte Carlo simulation. [Fig. 4](#) shows the probability density distribution for the metropolitan area planning policy. The distributions of the metropolitan area planning coefficients are insignificant and narrowly centered around zero. Therefore, the unobserved factors have almost no effect on the estimated results, ensuring the robustness of the baseline results.

### 4.4. Robust check

#### 4.4.1. Endogeneity mitigation

As the MADP depending on the government relationship and associate between city  $i$  and  $j$  can be endogenous, thus bias the OLS estimates. To solve this problem, we use the absolute value of the mean slope difference as an instrumental variable in our estimation. The difference of mean slope meets the exogenous requirement because the natural geographical factor was determined in early era, which is the exogenous shocks in the field of urban economics ([Combes & Gobillon, 2015](#); [Nunn & Puga, 2012](#)). Furthermore, *diff\_slope* is negatively correlated with *madv* and thus meeting the relevancy requirement. The rationale aligns with the work of [Nunn and Puga \(2012\)](#), which revealed that rugged terrains influence institutional development and local progress. Similarly, [Song and Chen \(2023\)](#) employed the degree of land surfaces as an instrumental variable due to its connection with informal institutions. When viewed in tandem, these findings imply that natural geography can shape institutions or governmental development strategies, such as MADP. This underscores the evident impact of rugged terrains on a government's regional development planning.

In column (1,2) of [Table 3](#), we reported regression results of difference of the mean slope between city  $i$  and  $j$  as IV instrument for *madv*. The coefficients in column (1,2) of [Table 3](#) are much larger than the coefficients in column (1,2) of [Table 3](#) because the endogeneity biased the estimates downward as we discussed in last section. When we include control variables,  $City_i \times Year$  fixed effects and  $City_j \times Year$  fixed effects in the instrumental variable regression, the coefficient of *madv*, which is city  $i$  and  $j$  in, is 1.2828 for the measure of the extensive margin, and 11.9508 for the measure of the intensive margin. Since the implication of MADP, increasing 1.28 times for extensive margin, i.e., the firm number of invest at city-pair level, and 11.95 times for intensive margin, i.e., total amount of investment at city-pair level.

The result for the first stage is also presented in column (3) of [Table 3](#), which implies that the instrument variable is significantly related with the endogenous variable. The Kleibergen-Paap rk LM statistic and Kleibergen-Paap rk Wald F statistic are shown in [Table 3](#), which is  $>10$ , indicating that our instrument passes the test for weak identification and underidentification comfortably.

From [Table 3](#), we can see that MADP can significantly facilitate inter-regional investment, especially when we correct for the endogenous issue regarding the construction of MADPs. The impacts of MADP are reflected on both extensive margin and intensive margin.

#### 4.4.2. Executive's hometown investments

To mitigate potential biases that may arise from familial or sentimental factors influencing investment decisions, we have conducted an analysis using a more restricted sample, excluding investments made by chairmen and general managers in their hometowns. The regression results are reported in Columns (1,2) of [Table 4](#). The significance and signs of MADP remain consistent with the benchmark regressions.

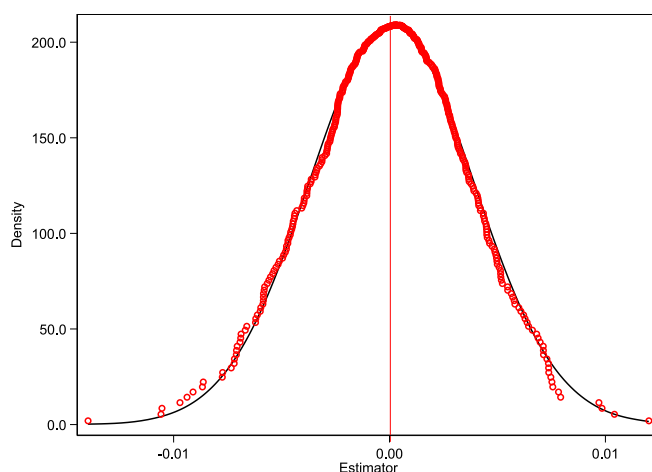


Fig. 4. Placebo test of MADP.



**Table 3**  
Endogeneity mitigation.

| Variables                            | (1)                   | (2)                    | (3)                                            |
|--------------------------------------|-----------------------|------------------------|------------------------------------------------|
|                                      | <i>lnfrm</i>          | <i>lncap</i>           | <i>madp</i>                                    |
| <i>madp</i>                          | 1.2828***<br>(0.3052) | 11.9508***<br>(2.8085) |                                                |
| <i>_cons</i>                         |                       |                        | −0.0021**<br>(0.0008)<br>0.2289***<br>(0.0573) |
| <i>Control variables</i>             | YES                   | YES                    | YES                                            |
| <i>City<sub>i</sub> × Year FE</i>    | YES                   | YES                    | YES                                            |
| <i>City<sub>j</sub> × Year FE</i>    | YES                   | YES                    | YES                                            |
| <i>N</i>                             | 1,507,772             | 1,507,761              | 1,507,772                                      |
| Kleibergen-Paap rk Wald LM statistic |                       |                        | 8.836                                          |
| Kleibergen-Paap rk F statistic       |                       |                        | 17.499                                         |

Note: (i) Robust standard errors, clustered at city pair and year level, are reported in brackets. (ii) \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ . (iii) Columns (1) and (2) show the Kleibergen-Paap rk LM statistic for underidentification test, and Kleibergen-Paap rk Wald F statistic for weak identification test.

**Table 4**  
Robust result.

|                                   | <i>lnfrm</i>                             | <i>lncap</i>          | <i>lnfrm</i>                                   | <i>lncap</i>          | <i>frm</i>            |
|-----------------------------------|------------------------------------------|-----------------------|------------------------------------------------|-----------------------|-----------------------|
|                                   | (1)                                      | (2)                   | (3)                                            | (4)                   | (5)                   |
| Variables                         | Without executive's hometown investments |                       | Without municipalities and provincial capitals |                       | PPML                  |
| <i>madp</i>                       | 0.0756***<br>(0.0213)                    | 0.5935***<br>(0.1545) | 0.0522***<br>(0.0165)                          | 0.7784<br>(0.2462)    | 0.3569***<br>(0.1383) |
| <i>_cons</i>                      | 0.0604***<br>(0.0018)                    | 0.5059***<br>(0.0130) | 0.0384***<br>(0.0013)                          | 0.3340***<br>(0.0094) | 4.2210***<br>(0.0901) |
| <i>Control variables</i>          | YES                                      | YES                   | YES                                            | YES                   | YES                   |
| <i>City<sub>i</sub> × Year FE</i> | YES                                      | YES                   | YES                                            | YES                   | YES                   |
| <i>City<sub>j</sub> × Year FE</i> | YES                                      | YES                   | YES                                            | YES                   | YES                   |
| <i>N</i>                          | 1,507,772                                | 1,507,761             | 1,346,936                                      | 1,346,929             | 1,298,220             |
| <i>Adj. R<sup>2</sup></i>         | 0.3542                                   | 0.1792                | 0.2843                                         | 0.1274                |                       |

Note: Column (1,2) represents the dependent variable, which excludes observations of chairmen and general managers' hometown investments and aggregates them into the inter-regional investment of the city pair. In column (3,4), inter-regional investments in cities with municipalities and provincial capitals as capital outflows are excluded. Column(5) is regressed using the Poisson Pseudo Maximum Likelihood (PPML) model.

#### 4.4.3. Municipalities and provincial capitals

To account for potential interference from city-level factors and the resource allocation capacity of provincial capitals and municipalities on the identification of MADP policies, we have conducted robustness tests on sub-samples, excluding relevant city pairs involving provincial capitals and municipalities. The results, as shown in column (3,4) the Table 4. The estimated coefficient of MADP was significantly positive, demonstrate that MADP indeed has a positive impact on inter-regional investment between non-provincial capitals and municipalities, indicating that municipalities and provincial capitals did not affect our estimation results.

#### 4.4.4. Zero invest value

In the baseline regression, one dependent variable is the natural logarithm of the number of firms investing between cities, with the addition of 1. To deal with zero investment quantity values, we also conduct a regression using the Poisson pseudo-maximum likelihood (PPML) method. The estimation results are shown in column (5) of Table 3. The estimated coefficient of policy was significantly positive, again demonstrating the robustness of the baseline estimation results.

## 5. Further analyses

### 5.1. Mechanism analysis

How do metropolitan planning policies boost the intensive and extensive margins of capital flow between cities? We study the institutional distance, market segmentation, and HSR connectivity indicated in Section 2.2 in light of institutional barriers.

#### 5.1.1. Institution distance

One significant impediment to the flow of capital across different regions is the institutional distance stemming from administrative boundaries (Aleksynska & Havrylych, 2013; Pellegrino, Spolaore, & Wacziarg, 2021). To evaluate whether MADP may lower institutional distance and encourage firms to invest in other places, this paper uses institutional distance (*insdis*) to estimate effect of

MADP. We adopt the method proposed by Chao and Kumar (2010) and Bae and Salomon (2010) to measure institutional distance, calculating  $insdis_{ijt} = \left[ \sum_{k=1}^5 |I_{kit} - I_{kjt}| \right] / 5$ . Here,  $I_{kit}$  represents the five types of market indices derived from China's Provincial Marketization Index Report. The Index comprises five specific indicators that reflect various aspects of institutional differences in marketization across regions, which is commonly used as institutional distance (Bae & Salomon, 2010).

The findings in Column (1) of Table 5 reveal that the coefficient of MADP holds the potential to effectively reduce institutional distance across regions. This underscores the pivotal role of MADP in reducing institutional distance and promoting enhanced enterprise investments between cities, consistent with prior literature findings (Meyer, 2005). Furthermore, the observed reduction in institutional distance aligns with the broader discussion on its positive influence on inter-regional investments, as extensively discussed in prior literature (Alfaro, Kalemli-Ozcan, & Volosovych, 2008; Lucas, 1990).

### 5.1.2. Market segmentation

The inter-city investment of a firm is the reallocation of stock capital as well as the micro-process of inter-regional capital flow. Even so, stock capital has significant local stickiness. Institutional obstacles and market segmentation linked to local government competition impede inter-regional capital flows, particularly in the context of China's decentralized economy (Song & Chen, 2023; Z. Chen, Haynes, Zhou, & Dai, 2019).

In this paper, eight commodity price indices are selected to calculate the city-level market segmentation index (*mktseg*) to represent the degree of market segmentation between different cities for mechanism analysis. Column (2) of Table 5 reveals a notable and statistically significant negative impact of MADP on market segmentation. This indicates that MADP plays a crucial role in diminishing market segmentation between cities. According to existing literature, a reduction in market segmentation tends to incentivize enterprises to consider investments in different geographical locations (Barca, McCann, & Rodríguez-Pose, 2012; Moschieri et al., 2014).

### 5.1.3. HSR connectivity

Transportation infrastructure reflects government cooperation across regions. However, challenges persist in connecting different areas, like providing public transportation at border regions, which poses difficulties for government collaboration. Integrating and developing public transportation across regions is vital to MADP and showcases intergovernmental cooperation. Leveraging high-speed railway data, we construct an intercity direct rail network. The dummy variable *hsrcon* is set to 1 if cities share a high-speed rail line, and 0 if not.

We examined the impact of MADP on regional government cooperation in transportation infrastructure, as evidenced in column 3 of Table 5. The results reveal a significantly positive coefficient for MADP, signifying its effective role in advancing high-speed rail connectivity between regions and bolstering regional accessibility. This enhanced accessibility leads to lowered information communication and trade expenses, thereby notably stimulating cross-regional capital flow (Lin, 2017; Pellegrino et al., 2021).

## 5.2. Heterogeneity analysis

This paper employs Boomhower (2019) method of heterogeneity analysis, which entails the interaction of two categorical dummy variables within the regression framework. The objective of this approach is to assess the impacts of MADP across distinct groups within the same regression, allowing for a direct observation and comparison of the magnitude of MADP's effects among these groups. Additionally, traditional interaction tests have also been conducted and are elaborated upon in the appendix due to space constraints. Subsequent interaction tests involving the same classification variables have been undertaken for consistency and are reported

**Table 5**  
Mechanism analysis.

| Variables                         | (1)                   | (2)                   | (3)                   |
|-----------------------------------|-----------------------|-----------------------|-----------------------|
|                                   | <i>insdis</i>         | <i>mktseg</i>         | <i>hsrcon</i>         |
| <i>madp</i>                       | −0.1842**<br>(0.0654) | −0.0001*<br>(0.0000)  | 0.0101***<br>(0.0034) |
| <i>_cons</i>                      | 1.9692***<br>(0.0054) | 0.0008***<br>(0.0000) | 0.0023***<br>(0.0003) |
| Control variables                 | YES                   | YES                   | YES                   |
| <i>City<sub>i</sub> × Year FE</i> | YES                   | YES                   | YES                   |
| <i>City<sub>j</sub> × Year FE</i> | YES                   | YES                   | YES                   |
| <i>N</i>                          | 1,507,772             | 945,532               | 1,507,772             |
| Adj. <i>R</i> <sup>2</sup>        | 0.4356                | 0.7048                | 0.1545                |

Note: Institutional distance in column (1) is calculated as follows:  $insdis_{ijt} = \left[ \sum_{k=1}^5 |I_{kit} - I_{kjt}| \right] / 5$ , where  $I_{kit}$  is the 5 kinds of market index from China's Provincial Marketization Index Report; There is a lack of market segmentation data at the municipal level in the column (2). The statistical caliber of the eight categories of urban commodity price data has changed since 2016. Therefore, this article only uses the market segmentation index *mktseg* calculated from 2001 to 2015, resulting in inconsistent observation.

accordingly.

### 5.2.1. Heterogeneity analysis based on application of bigdata

Heterogeneity exists in the influence of MADP across regions characterized by varying degrees of big data application (Hellwig, 2022), resulting in diverse policy responses to MADP. This paper employs the number of Internet broadband access users as a metric for big data application indicators (Niu, Jin, Wang, & Zhou, 2022), categorized by median values, to discern the origins and trajectories of capital flows influenced by the extent of big data integration. The findings are detailed in Table 6. In Panel A, we compare and assess the impact of MADP on capital outflows (source cities) across different levels of big data application. The analysis reveals that MADP significantly stimulates local capital outflows in regions with robust big data application foundations. In Panel B, we scrutinize the effect of MADP on capital inflows (destination cities) in varying degrees of big data application. Here, we observe that MADP plays a substantial role in fostering local capital inflows in regions with strong big data application bases. Conversely, in regions with weaker big data application foundations, MADP exhibits a negative influence on capital inflows. This contrast underscores MADP's notable ability to facilitate inter-regional capital flows in regions with well-established big data application frameworks, while this effect is not evident in regions with less developed big data application infrastructure.

### 5.2.2. Heterogeneity analysis based on border

Taking into account the presence of border effects, the potential integration impact of MADP may face challenges in exerting its full influence, thus impeding cooperation and coordination across regions. This paper examines the influence of provincial administrative boundaries on the cross-border effectiveness of MADP, and the findings are detailed in Table 7. Notably, regardless of the presence of boundaries, MADP demonstrates a significant capacity to enhance the flow of capital circulation. Specifically, in cities devoid of boundaries, the impact of MADP on the sample exhibits a high level of significance at the 1% level, while in cities with boundaries, the coefficient of MADP remains statistically significant at the 5% level. The underlying rationale could be attributed to MADP serving as instruments of regional integration. By fostering governmental collaboration and mitigating the influence of administrative boundaries, these plannings contribute to overcoming impediments to resource flow. Consequently, this policy dynamic facilitates the movement of capital and other critical factors across various regions.

### 5.2.3. Heterogeneity analysis based on the TFP

Differences in policies stemming from various levels of total factor productivity (TFP) can be observed in relation to MADP, a vital policy driving balanced regional growth and economic convergence among cities. Employing Stochastic Frontier Analysis (SFA) to compute regional total factor productivity (Battese & Coelli, 1992; Belotti, Daidone, Ilardi, & Atella, 2013), this paper categorizes virtual variables by annual medians and presents the findings in Table 8. Notably, both the coefficient of MADP on capital flow in high and low TFP cities remains statistically significant at the 1% level. However, its impact on capital inflows and outflows is more

**Table 6**  
Heterogeneity analysis based on application of bigdata.

| Variables                                | (1)                            | (2)                           |
|------------------------------------------|--------------------------------|-------------------------------|
|                                          | <i>lnfnm</i>                   | <i>lncap</i>                  |
| Panel A: Source cities                   |                                |                               |
| <i>nobigdata</i> × <i>madp</i>           | 0.0056<br>(0.0071)             | 0.0643<br>(0.0753)            |
| <i>bigdata</i> × <i>madp</i>             | 0.1156***<br>(0.0331)          | 0.8960***<br>(0.2366)         |
| <i>_cons</i>                             | 0.0639***<br>(0.0019)          | 0.5486***<br>(0.0137)         |
| Control variables                        | YES                            | YES                           |
| <i>City<sub>i</sub></i> × <i>Year FE</i> | YES                            | YES                           |
| <i>City<sub>j</sub></i> × <i>Year FE</i> | YES                            | YES                           |
| <i>N</i>                                 | 1,378,169                      | 1,378,158                     |
| Adj. <i>R</i> <sup>2</sup>               | 0.3619                         | 0.1848                        |
| Panel B: Destination cities              |                                |                               |
| <i>nobigdata</i> × <i>madp</i>           | 0.3573<br>−0.0179*<br>(0.0094) | 0.1789<br>−0.0541<br>(0.0850) |
| <i>bigdata</i> × <i>madp</i>             | 0.1316***<br>(0.0329)          | 0.9832***<br>(0.2255)         |
| <i>_cons</i>                             | 0.0638***<br>(0.0018)          | 0.5479***<br>(0.0132)         |
| Control variables                        | YES                            | YES                           |
| <i>City<sub>i</sub></i> × <i>Year FE</i> | YES                            | YES                           |
| <i>City<sub>j</sub></i> × <i>Year FE</i> | YES                            | YES                           |
| <i>N</i>                                 | 1,378,169                      | 1,378,158                     |
| Adj. <i>R</i> <sup>2</sup>               | 0.3624                         | 0.1851                        |

Note: Due to the missing number of Internet broadband access users in the China City Statistical Yearbook, the observations is inconsistent with other regressions.

**Table 7**  
Heterogeneity analysis based on border.

| Variables                          | (1)                   | (2)                   |
|------------------------------------|-----------------------|-----------------------|
|                                    | <i>lnfnm</i>          | <i>lnicap</i>         |
| <i>withoutborder</i> × <i>madp</i> | 0.6478***<br>(0.0923) | 5.5062***<br>(0.6496) |
| <i>border</i> × <i>madp</i>        | 0.0432**<br>(0.0182)  | 0.3160**<br>(0.1354)  |
| <i>_cons</i>                       | 0.0602***<br>(0.0018) | 0.5063***<br>(0.0128) |
| Control variables                  | YES                   | YES                   |
| <i>City<sub>i</sub></i> × Year FE  | YES                   | YES                   |
| <i>City<sub>j</sub></i> × Year FE  | YES                   | YES                   |
| <i>N</i>                           | 1,507,772             | 1,507,761             |
| Adj. <i>R</i> <sup>2</sup>         | 0.3685                | 0.1899                |

**Table 8**  
Heterogeneity analysis based on the TFP.

| Variables                         | (1)                   | (2)                   |
|-----------------------------------|-----------------------|-----------------------|
|                                   | <i>lnfnm</i>          | <i>lnicap</i>         |
| Panel A: Source city              |                       |                       |
| <i>lowtfp</i> × <i>madp</i>       | 0.0787***<br>(0.0225) | 0.6216***<br>(0.1636) |
| <i>hightfp</i> × <i>madp</i>      | 0.0728***<br>(0.0207) | 0.5695***<br>(0.1480) |
| <i>_cons</i>                      | 0.0604***<br>(0.0018) | 0.5076***<br>(0.0130) |
| Control variables                 | YES                   | YES                   |
| <i>City<sub>i</sub></i> × Year FE | YES                   | YES                   |
| <i>City<sub>j</sub></i> × Year FE | YES                   | YES                   |
| <i>N</i>                          | 1,503,216             | 1,503,205             |
| Adj. <i>R</i> <sup>2</sup>        | 0.3538                | 0.1794                |
| Panel B: Destination city         |                       |                       |
| <i>lowtfp</i> × <i>madp</i>       | 0.0860***<br>(0.0239) | 0.6598***<br>(0.1695) |
| <i>hightfp</i> × <i>madp</i>      | 0.0661***<br>(0.0192) | 0.5328***<br>(0.1433) |
| <i>_cons</i>                      | 0.0603***<br>(0.0018) | 0.5071***<br>(0.0130) |
| Control variables                 | YES                   | YES                   |
| <i>City<sub>i</sub></i> × Year FE | YES                   | YES                   |
| <i>City<sub>j</sub></i> × Year FE | YES                   | YES                   |
| <i>N</i>                          | 1,503,216             | 1,503,205             |
| Adj. <i>R</i> <sup>2</sup>        | 0.3543                | 0.1796                |

Note: Due to a minor amount of missing data in the input indicator of employment used for calculating TFP, the observations is inconsistent with other regressions.

pronounced in regions with lower TFP. Specifically, Panel A highlights a stronger role of MADP in promoting capital outflows from low TFP cities, with a higher MADP coefficient. Similarly, Panel B shows a heightened coefficient for capital inflows in low TFP cities. This could be attributed to effect of MADP to diminish communication costs and information barriers within TFP-undeveloped regions, along with targeted policies fostering substantial capital flow growth in low TFP regions.

## 6. Conclusions and implications

Using data on investments by listed companies and the scope of MADP from 1999 to 2019, this paper investigates effects of MADP on inter-regional investment and explores their intrinsic mechanisms in depth. Specifically, we arrive at three findings. Firstly, the adoption of MADP stimulates unhindered capital movement and amplifies both the intensive and extensive dimensions of corporate investments. Secondly, our mechanism analysis discloses that MADP promotes the capital flow by decreasing institutional distance, eliminating market segmentation, and enhancing the connectivity of transportation infrastructure. Thirdly, the influence of MADP on capital flows is more significant in cities endowed with robust big data application foundations, lower TFP levels, and in city pairs devoid of administrative borders.

This paper enriches the literature on capital flow motivations by exploring the effect of metropolitan area planning on capital flows. Existing literature on MADP has been enriched by studying the intensive and extensive margins of different capital flows.

The empirical findings indicate that metropolitan area planning has the capacity to mitigate the impact of institutional distance and market segmentation, while simultaneously enhancing transportation connectivity for capital flow. This highlights the importance of strategically advancing the spatial arrangement of MADP, with the goal of breaking down administrative barriers across institutional, trade, and spatial dimensions. By doing so, MADP can effectively catalyze inter-regional cooperation and coordination, further extending the realms of urbanization. This, in turn, facilitates the creation of a unified regional market, fostering economies of scale and amplifying the overall impact of MADP.

Furthermore, heterogeneity analysis reveals that MADP serves to stimulate capital flow particularly to cities boasting robust big data infrastructure but lower TFP. This highlights the pivotal role of MADP in fostering knowledge diffusion and fostering balanced development. By meticulously integrating factors like resource endowments, economic foundations, and industrial advancement across regions, strategically configuring the big data infrastructure, MADP emerges as a catalyst for propelling coordinate economic growth.

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## Data availability

Data will be made available on request.

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## Appendix A

**Table A1**

Cities covered by MADPs.

| MADPs                                            | Year | Cities                                                                                                   |
|--------------------------------------------------|------|----------------------------------------------------------------------------------------------------------|
| Changzhutan Metropolitan Area                    | 2000 | Changsha, Zhuzhou, Xiangtan                                                                              |
| Xuzhou Metropolitan Area                         | 2001 | Xuzhou, Lianyungang, Suqian; Suzhou and Huaibei; Zaozhuang, Jinling, Shangqiu                            |
| Nanjing Metropolitan Area                        | 2002 | Nanjing, Zhenjiang, Yangzhou, Ma'anshan, Chuzhou, Wuhu, Huai'an, Hefei                                   |
| Harbin Metropolitan Area                         | 2005 | Harbin, Suihua                                                                                           |
| Taiyuan Metropolitan Area                        | 2005 | Taiyuan, Jinzhong, Yangquan, Xinzhou, Lvliang                                                            |
| Shenyang Metropolitan Area                       | 2007 | Shenyang, Anshan, Fushun, Benxi, Yingkou, Fuxin, Liaoyang, Tieling                                       |
| Wuhan Metropolitan Area                          | 2007 | Wuhan, Huangshi, Ezhou, Xiaogan, Huanggang, Xianning                                                     |
| Guiyang Metropolitan Area                        | 2007 | Guiyang, Anshun                                                                                          |
| Hefei Metropolitan Area                          | 2008 | Hefei, Huainan, Bengbu, Chuzhou, Ma'anshan, Anqing, Wuhu, Tongling                                       |
| Xian Metropolitan Area                           | 2009 | Xi'an, Xianyang, Baoji, Tongchuan, Weinan, Shangluo, Yangling, Tianshui                                  |
| Capital Metropolitan Area                        | 2010 | Beijing, Tianjin, Shijiazhuang, Qinhuangdao, Tangshan, Langfang, Baoding, Cangzhou, Zhangjiakou, Chengde |
| Hangzhou Metropolitan Area                       | 2010 | Hangzhou, Huzhou, Jiaxing, Shaoxing                                                                      |
| Ningbo Metropolitan Area                         | 2010 | Ningbo, Zhoushan, Taizhou                                                                                |
| Suzhou Wuxi Changzhou Metropolitan Area          | 2011 | Shijiazhuang, Zhengding, Luquan, Luancheng, Gaocheng                                                     |
| Fuzhou Metropolitan Area                         | 2011 | Fuzhou, Ningde, Putian, Nanping                                                                          |
| Urumqi Metropolitan Area                         | 2012 | Ürümqi, Turpan                                                                                           |
| Lanzhou Metropolitan Area                        | 2013 | Lanzhou, Baiyin, Dingxi                                                                                  |
| Changchun Metropolitan Area                      | 2014 | Changchun, Jilin                                                                                         |
| Xining Metropolitan Area                         | 2015 | Xining, Haidong                                                                                          |
| Jinan Metropolitan Area                          | 2016 | Jinan, Dezhou, Tai'an, Binzhou, Tai'an, Zibo, Liaocheng                                                  |
| Qingdao Metropolitan Area                        | 2016 | Qingdao, Weifang and Yantai                                                                              |
| Guangzhou Metropolitan Area                      | 2016 | Guangzhou, Foshan, Shaoguan, Zhaoqing, Qingyuan and Yunfu.                                               |
| Hong Kong Shenzhen Metropolitan Area             | 2016 | Shenzhen, Heyuan, Huizhou, Shanwei, Dongguan                                                             |
| Pearl River Estuary West Coast Metropolitan Area | 2016 | Zhuhai, Zhongshan, Jiangmen, Yangjiang                                                                   |
| Yinchuan Metropolitan Area                       | 2017 | Yinchuan, Shizuishan, Wu Zhong                                                                           |
| Shanghai Metropolitan Area                       | 2017 | Suzhou, Wuxi, Changzhou, Nantong, Jiaxing, Ningbo, Zhoushan, Huzhou, Hangzhou                            |
| Dalian Metropolitan Area                         | 2018 | Dalian, Yingkou, Anshan, Dandong                                                                         |
| Hohhot Metropolitan Area                         | 2018 | Hohhot, Baotou, Ordos, Ulanqab                                                                           |
| Nanchang Metropolitan Area                       | 2018 | Nanchang, Jiujiang, Fuzhou, Yichun, Shangrao                                                             |

Note: Since our main focus is on inter-city investment relationships, we have excluded the metropolitan areas development planning of Shijiazhuang and Chongqing. This decision is based on the fact that these plans do not extend beyond the boundaries of the respective cities.

**Table A2**  
Heterogeneity analysis based on application of bigdata.

|                                   | (1)                   | (2)                   |
|-----------------------------------|-----------------------|-----------------------|
|                                   | <i>lnfrm</i>          | <i>lncap</i>          |
| Panel A: Source cities            |                       |                       |
| <i>madp</i>                       | 0.0056<br>(0.0071)    | 0.0643<br>(0.0753)    |
| <i>bigdata</i> × <i>madp</i>      | 0.1101***<br>(0.0349) | 0.8317***<br>(0.2552) |
| <i>_cons</i>                      | 0.0639***<br>(0.0019) | 0.5486***<br>(0.0136) |
| Control variables                 | YES                   | YES                   |
| <i>City<sub>i</sub></i> × Year FE | YES                   | YES                   |
| <i>City<sub>j</sub></i> × Year FE | YES                   | YES                   |
| <i>N</i>                          | 1,378,169             | 1,378,158             |
| Adj. <i>R</i> <sup>2</sup>        | 0.3573                | 0.1789                |
| Panel B: Destination cities       |                       |                       |
| <i>madp</i>                       | −0.0179*<br>(0.0094)  | −0.0541<br>(0.0850)   |
| <i>bigdata</i> × <i>madp</i>      | 0.1495***<br>(0.0364) | 1.0373***<br>(0.2419) |
| <i>_cons</i>                      | 0.0638***<br>(0.0018) | 0.5479***<br>(0.0130) |
| Control variables                 | YES                   | YES                   |
| <i>City<sub>i</sub></i> × Year FE | YES                   | YES                   |
| <i>City<sub>j</sub></i> × Year FE | YES                   | YES                   |
| <i>N</i>                          | 1,378,169             | 1,378,158             |
| Adj. <i>R</i> <sup>2</sup>        | 0.3578                | 0.1792                |

**Table A3**  
Heterogeneity analysis based on border.

|                                   | (1)                    | (2)                    |
|-----------------------------------|------------------------|------------------------|
|                                   | <i>lnfrm</i>           | <i>lncap</i>           |
| <i>border</i>                     | −0.2325***<br>(0.0306) | −2.2900***<br>(0.3179) |
| <i>madp</i>                       | 0.4186***<br>(0.0769)  | 3.2488***<br>(0.5621)  |
| <i>border</i> × <i>madp</i>       | −0.3700***<br>(0.0649) | −2.8798***<br>(0.4928) |
| <i>_cons</i>                      | 0.2835***<br>(0.0285)  | 2.7051***<br>(0.3010)  |
| Control variables                 | YES                    | YES                    |
| <i>City<sub>i</sub></i> × Year FE | YES                    | YES                    |
| <i>City<sub>j</sub></i> × Year FE | YES                    | YES                    |
| <i>N</i>                          | 1,507,772              | 1,507,761              |
| Adj. <i>R</i> <sup>2</sup>        | 0.3880                 | 0.2076                 |

**Table A4**  
Heterogeneity analysis based on TFP.

|                                   | (1)                   | (2)                   |
|-----------------------------------|-----------------------|-----------------------|
|                                   | <i>lnfrm</i>          | <i>lncap</i>          |
| Panel A: Source cities            |                       |                       |
| <i>madp</i>                       | 0.0787***<br>(0.0225) | 0.6216***<br>(0.1636) |
| <i>hightfp</i> × <i>madp</i>      | −0.0059<br>(0.0069)   | −0.0521<br>(0.0346)   |
| <i>_cons</i>                      | 0.0604***<br>(0.0018) | 0.5076***<br>(0.0130) |
| Control variables                 | YES                   | YES                   |
| <i>City<sub>i</sub></i> × Year FE | YES                   | YES                   |
| <i>City<sub>j</sub></i> × Year FE | YES                   | YES                   |
| <i>N</i>                          | 1,503,216             | 1,503,205             |
| Adj. <i>R</i> <sup>2</sup>        | 0.3538                | 0.1794                |
| Panel B: Destination cities       |                       |                       |
| <i>madp</i>                       | 0.0860***<br>(0.0239) | 0.6598***<br>(0.1695) |

(continued on next page)



Table A4 (continued)

|                                          | (1)                    | (2)                   |
|------------------------------------------|------------------------|-----------------------|
|                                          | <i>lnfnm</i>           | <i>lnicap</i>         |
| <i>hightfp</i> × <i>madp</i>             | −0.0199***<br>(0.0068) | −0.1270**<br>(0.0457) |
| <i>_cons</i>                             | 0.0603***<br>(0.0018)  | 0.5071***<br>(0.0130) |
| Control variables                        | YES                    | YES                   |
| <i>City<sub>i</sub></i> × <i>Year FE</i> | YES                    | YES                   |
| <i>City<sub>j</sub></i> × <i>Year FE</i> | YES                    | YES                   |
| <i>N</i>                                 | 1,503,216              | 1,503,205             |
| Adj. <i>R</i> <sup>2</sup>               | 0.3543                 | 0.1796                |

## Appendix B

To measure the regional TFP, we begin by setting the regional Stochastic Frontier Production (SFA) function as follows

$$Y_{it} = f(K_{it}, L_{it}, t) \exp(v_{it} - u_{it}) \quad (\text{A.1})$$

Taking the natural logarithm on both sides yields:

$$\ln Y_{it} = \ln f(K_{it}, L_{it}, t) + v_{it} - u_{it} \quad (\text{A.2})$$

where  $Y_{it}$  represents the total output of city  $i$  in year  $t$ , and  $f(K_{it}, L_{it}, t)$  is its frontier production function, commonly in Cobb-Douglas form.  $K_{it}$  denotes the input of capital,  $L_{it}$  is the input of labor, and  $(v_{it} - u_{it})$  represents the composite error term.

Due to the numerous assumptions and limitations associated with the use of the Cobb-Douglas function, and since the transcendental logarithmic function relaxes the assumption of neutral technological progress, it exhibits greater inclusivity compared to the Cobb-Douglas. Consequently, this paper employs a time-varying transcendental logarithmic production function model to measure changes in technological progress and technical efficiency as follows:

$$\ln Y_{it} = \beta_0 + \beta_K \ln K_{it} + \beta_L \ln L_{it} + \frac{1}{2} \beta_{KK} (\ln K_{it})^2 + \frac{1}{2} \beta_{LL} (\ln L_{it})^2 + \beta_{KL} \ln K_{it} \ln L_{it} + \beta_{Kt} t \ln K_{it} + \beta_{Lt} t \ln L_{it} + \beta_t t + \frac{1}{2} \beta_{tt} t^2 + v_{it} - u_{it} \quad (\text{A.3})$$

where  $Y_{it}$  represents the GDP of city  $i$  in year  $t$ ,  $K_{it}$  denotes the stock of fixed assets,  $L_{it}$  is the number of employees. The elasticity coefficient,  $\beta_i$ , while  $t$  represents the time trend variable that responds to technological change effects.  $v_{it}$  represents the impact of random shocks,  $v_{it} \sim N(0, \sigma_v^2)$ .  $u_{it}$  signifies the influence of technology efficiency with random disturbances removed, where  $\exp(-u_{it}) \in [0, 1]$ . This paper adopts the assumptions for  $u_{it}$  and  $e_{it}$  based on Battese and Coelli (1992), as follows:

$$u_{it} \sim N^+(\mu, \sigma_u^2) \quad (\text{A.4})$$

$$u_{it} = u_i \exp[-\eta(t - T)] \quad (\text{A.5})$$

$$e_{it} = v_{it} - \eta_{it} u_{it} \quad (\text{A.6})$$

$$\gamma = \frac{\sigma_u^2}{\sigma_v^2 + \sigma_u^2}, \sigma^2 = \sigma_v^2 + \sigma_u^2, u_i \sim N(\mu, \sigma_u^2) \quad (\text{A.7})$$

Based on Eq. (A.3), we simultaneously estimate all parameters using maximum likelihood estimation to determine the technical efficiency:

$$TE_{it} = \exp(-u_{it}) \quad (\text{A.8})$$

Combining the setting of the Malmquist index distance function, the change in TFP for city  $i$  between time  $t$  and  $t + 1$  can be calculated, yielding a measure of technological change:

$$TEC_i^{t,t+1} = \frac{TE_{i,t+1}}{TE_{it}} \quad (\text{A.9})$$

Using the method proposed by Coelli, Prasada Rao, O'Donnell, and Battese (2005) for calculating technological change, we apply the translog production function to derive partial derivatives of output for each city between two time periods, resulting in the calculation of the technological change index.

$$TPC_i^{t,t+1} = \exp \left[ \frac{1}{2} \left( \frac{\partial \ln Y_{i,t+1}}{\partial (t+1)} + \frac{\partial \ln Y_{it}}{\partial t} \right) \right] \quad (\text{A.10})$$

Finally, the product of the indexes of technological efficiency change and technological progress change is obtained:

$$TFP_i^{t,t+1} = TEC_i^{t,t+1} \times TPC_i^{t,t+1} \quad (\text{A.11})$$

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